

Government First Grade College and PG Centre, Dharwad



DEPARTMENT OF COMPUTER SCIENCE

SYNOPSIS FOR BCA SEMESTER PROJECT 2025-26

**Title of the Project : Cloud-Based Knowledge Repository and Digital
Resources Management System**

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2. Objective of the Project:

- **Develop a centralized digital repository:** To create a cloud-based platform for storing, managing, and accessing large volumes of knowledge and digital resources, including documents, multimedia files, and structured data.
 - **Ensure efficient retrieval:** Enable users to quickly search and retrieve documents using advanced indexing, metadata, and search algorithms.
 - **Support collaboration:** Allow users to upload, edit, annotate, and collaborate on documents in real-time.
 - **Scalable and secure infrastructure:** Leverage cloud infrastructure to ensure scalability, security, and accessibility for a global user base.
 - **Improve knowledge management:** Provide an effective solution for organizations to manage their internal knowledge and resources.
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3. Input of the Project:

- **User Input:** Users will upload documents, files, or data (e.g., PDFs, Word documents, multimedia files, etc.) to the cloud repository.
 - **Metadata:** Each resource will be tagged with metadata such as title, category, keywords, and description to facilitate search and retrieval.
 - **Search Queries:** Users input search terms or filters to retrieve documents from the repository.
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4. Output of the Project:

- **Document Retrieval:** Users will receive a list of relevant documents, reports, or resources based on their search query.
- **Collaboration Features:** Real-time updates when documents are edited, annotated, or shared among users.
- **Recommendations:** The system will recommend documents based on user behavior, preferences, and past searches.
- **Analytics Reports:** User activity and content analytics to track engagement, popular documents, and system usage patterns.

5. Process Logic:

- **Data Upload and Categorization:**
 - Users upload documents or files to the cloud repository. Metadata is automatically extracted or manually added to categorize and tag resources.
 - **Metadata Indexing:**
 - The system indexes the metadata (e.g., keywords, document titles, descriptions) and stores them in a structured database for fast retrieval.
 - **Content Storage:**
 - Documents and resources are stored in cloud object storage (e.g., AWS S3 or Google Cloud Storage) for scalability and durability.
 - **Search Functionality:**
 - The user can search using keywords or filters (such as category, date, or tags). The search engine (e.g., Elasticsearch or Solr) retrieves the most relevant documents.
 - **Collaboration and Versioning:**
 - Documents are versioned, and real-time collaboration is enabled using tools like Google Docs API or custom-built solutions.
 - **Access Control:**
 - Role-based access controls are implemented to restrict document access based on user permissions (read-only, edit, or admin).
 - **Recommendation System:**
 - A recommendation engine suggests documents based on user interactions, past searches, and content popularity.
 - **Analytics and Reporting:**
 - User activities and content performance are tracked and visualized using analytics tools, helping administrators improve the system.
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6. Tools / Platform / Languages:

- **Cloud Platforms:** Amazon Web Services (AWS), Google Cloud, Microsoft Azure.
 - **Storage Solutions:** AWS S3, Google Cloud Storage, Azure Blob Storage.
 - **Programming Language:** Python
 - **Framework:** Flask
 - **Database:** MySQL
 - **Frontend:** React.js, HTML, CSS
 - **Libraries Used:** NumPy
 - **Scikit-learn**
 - **Axios**
 - **PyMySQL**
 - **Platform:** Windows / Web Browser
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7.Are you doing this project for any Industry / Client?

Answer: No

8. Duration of the Project:

- **Estimated Duration:** 2 months
 - **Project Schedule**
 - **Phase 1:** Requirement Analysis — 10 Hours
 - **Phase 2:** System Design (UI & DB) — 15 Hours
 - **Phase 3:** Implementation (Coding) — 45 Hours
 - **Phase 4:** Testing & Debugging — 12 Hours
 - **Phase 5:** Deployment & Documentation — 8 Hours
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9. Limitations:

- **Dependency on Internet Access:** The system requires a stable internet connection for cloud access.
 - **Initial Learning Curve:** Users may face a slight learning curve in understanding how to upload and manage documents effectively.
 - **Cloud Service Costs:** Long-term operational costs depend on cloud service usage and scale.
 - **Integration Challenges:** Difficulty in integrating with legacy systems or proprietary formats may arise.
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10. Scope / Future Applications:

- **Multilingual Support:** Extend the system to support multiple languages for global organizations.
 - **AI and Machine Learning Integration:** Use AI models to automate document categorization, tagging, and content summarization.
 - **Mobile App Support:** Create mobile applications for users to access and manage knowledge resources on the go.
 - **Integration with Third-Party Systems:** Integrate with popular systems like **Learning Management Systems (LMS)**, **Customer Relationship Management (CRM)**, and **Enterprise Resource Planning (ERP)** solutions.
 - **Data Analytics & Business Intelligence:** Implement more advanced analytics, including trends and content consumption patterns for more informed decision-making.
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